

Accounting Firm Management & Financial Record Database

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I. PROJECT DETAILS

Project Title: Accounting Firm Management & Financial Record Database

Team Name: KQD

Team Member:

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II. PROBLEM STATEMENT

Modern accounting firms handle large amounts of data, including assets, liabilities, expenses, transactions, and more. In order to securely and efficiently handle these records, the use of relational databases are necessary, as they support human friendly relational queries and structure. In comparison with the previously used excel sheets, relational databases handle common errors such as data inconsistencies, redundancies, and data integrity. Additionally, in the specific case of accounting and finance firms, they will support modularity in multiple relations, addressing specific datasets such as transactions, expenses, departments, users, and more. Through the BCNF form, we will be able to create relations, reducing anomalies and inconsistencies, allowing our users to reduce manual work, streamline common database operations such as querying or updating, and ensuring efficient dataset interactions.

III. TARGET USER

This database is targeted towards Accountants and Financial Analysts, to assist with tracking/recording user information such as finances, assets, expenses, transactions, and more. This will allow professionals to analyze financial trends, maintain accurate data, and monitor departmental performance, streamlining manual work and preventing data errors with redundancies.

IV. E/R DIAGRAM

Refer to Fig 1., for the E/R Diagram

V. RELATIONS AND THEIR ATTRIBUTES

A. Departments Table

1) *department_id*: This attribute will serve as the Primary Key. The purpose of this attribute is to provide a unique ID for each department with INT as its datatype. Department ID will not have a default value or a NULL value; rather, it will start with 1 or the next available number that is not used in the relation.

2) *department_name*: This attribute will not serve as the Primary Key or the Foreign Key. The purpose of this attribute is to provide a name for the department with VARCHAR(50) as its datatype. Department Name will not have a default value or a NULL value.

3) *manager_id*: This attribute will serve as the Foreign Key. The purpose of this attribute is to assign a user as the manager of the department with INT as its datatype. Manager ID will not have a default value or a NULL value. If the primary

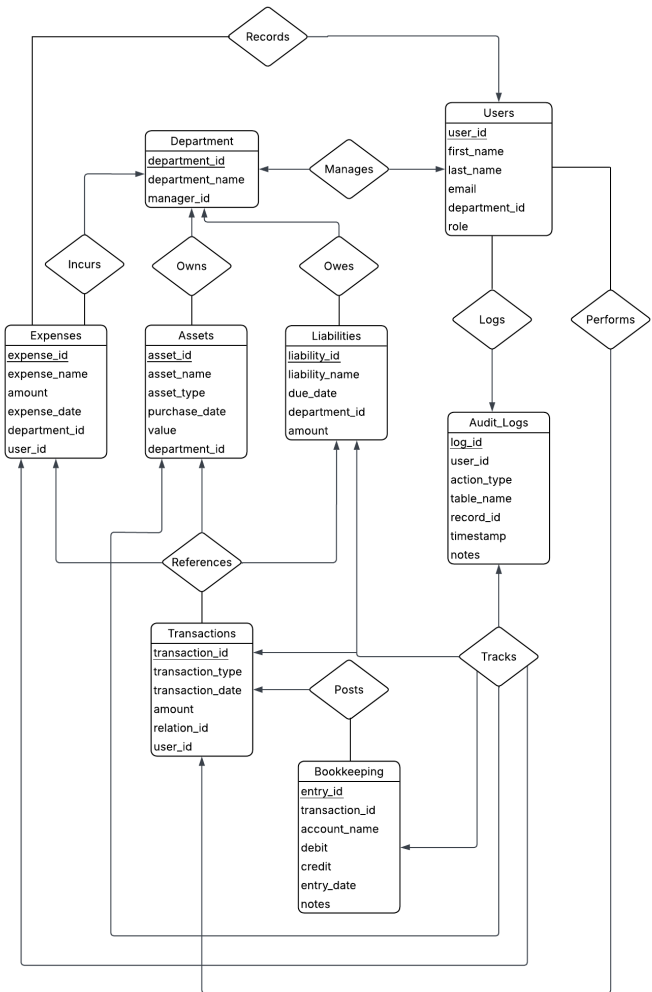


Fig. 1. E/R Diagram

key is deleted, DELETE CASCADE will be performed on the relation.

B. Users Table

1) *user_id*: This attribute will serve as the Primary Key. The purpose of this attribute is to provide a unique ID for each member with INT as its datatype. User ID will not have a default value or a NULL value; rather, it will start with 1 or the next available number that is not used in the relation.

2) *first_name*: This attribute will not serve as the Primary Key or the Foreign Key. The purpose of this attribute is to list the first name of the member, which will be a VARCHAR(100) as its datatype. First Name will not have a default value or a NULL value.

3) *last_name*: This attribute will not serve as the Primary Key or the Foreign Key. The purpose of this attribute is to list the last name of the member, which will be a VARCHAR(100) as its datatype. Last Name will not have a default value or a NULL value.

4) *email*: This attribute will not serve as the Primary Key or the Foreign Key. The purpose of this attribute is to list the member's email, which will be a VARCHAR(200) as its datatype. If an email is not provided, it will be set as NULL.

5) *department_id*: This attribute will serve as the Foreign Key. The purpose of this attribute is to link which department the member belongs to, and it will be an INT as its datatype. Department ID will not have a default value or a NULL value. If the primary key is deleted, DELETE CASCADE will be performed on the relation.

6) *role*: This attribute will not serve as the Primary Key or the Foreign Key. The purpose of this attribute is to provide a role for the member, which will be a VARCHAR(50) as its datatype. If a role is not provided, it will be set as NULL.

C. Assets Table

1) *asset_id*: This attribute will serve as the Primary Key. The purpose of this attribute is to provide a unique ID for each asset with INT as its datatype. Asset ID will not have a default value or a NULL value; rather, it will start with 1 or the next available number that is not used in the relation.

2) *asset_name*: This attribute will not serve as the Primary Key or the Foreign Key. The purpose of this attribute is to provide a name for the asset, which will be a VARCHAR(100) as its datatype. Asset Name will not have a default or NULL value.

3) *asset_type*: This attribute will not serve as the Primary Key or the Foreign Key. The purpose of this attribute is to provide a classification for the asset, which will be a VARCHAR(50) as its datatype. Asset Type will not have a default or NULL value.

4) *purchase_date*: This attribute will not serve as the Primary Key or the Foreign Key. The purpose of this attribute is to provide the purchase date of the asset, which will be a DATETIME as its datatype. Purchase Date will not have a default or NULL value.

5) *value*: This attribute will not serve as the Primary Key or the Foreign Key. The purpose of this attribute is to provide the value of the asset, which will be a DECIMAL as its datatype. Value will not have a default or NULL value.

6) *department_id*: This attribute will serve as the Foreign Key. The purpose of this attribute is to link which department the asset belongs to, and it will be an INT as its datatype. Department ID will not have a default value or a NULL value. If the primary key is deleted, DELETE CASCADE will be performed on the relation.

D. Liabilities Table

1) *liability_id*: This attribute will serve as the Primary Key. The purpose of this attribute is to provide a unique ID for each liability with INT as its datatype. Liability ID will not have a default value or a NULL value; rather, it will start with 1 or the next available number that is not used in the relation.

2) *liability_name*: This attribute will not serve as the Primary Key or the Foreign Key. The purpose of this attribute is to provide a name for the liability, which will be a VARCHAR(100) as its datatype. Liability Name will not have a default or NULL value.

3) *amount*: This attribute will not serve as the Primary Key or the Foreign Key. The purpose of this attribute is to provide the amount of the liability, which will be a DECIMAL as its datatype. Amount will not have a default or NULL value.

4) *due_date*: This attribute will not serve as the Primary Key or the Foreign Key. The purpose of this attribute is to provide the expiration date of the liability, which will be a DATETIME as its datatype. Due Date will not have a default or NULL value.

5) *department_id*: This attribute will serve as the Foreign Key. The purpose of this attribute is to link which department the liability belongs to, and it will be an INT as its datatype. Department ID will not have a default value or a NULL value. If the primary key is deleted, DELETE CASCADE will be performed on the relation.

E. Expenses Table

1) *expense_id*: This attribute will serve as the Primary Key. The purpose of this attribute is to provide a unique ID for each expense with INT as its datatype. Expense ID will not have a default value or a NULL value; rather, it will start with 1 or the next available number that is not used in the relation.

2) *expense_name*: This attribute will not serve as the Primary Key or the Foreign Key. The purpose of this attribute is to provide a name for the expense, which will be a VARCHAR(50) as its datatype. Expense Name will not have a default or NULL value.

3) *amount*: This attribute will not serve as the Primary Key or the Foreign Key. The purpose of this attribute is to provide the amount for the expense, which will be a DECIMAL as its datatype. Amount will not have a default or NULL value.

4) *department_id*: This attribute will serve as the Foreign Key. The purpose of this attribute is to link which department the expense belongs to, and it will be an INT as its datatype. Department ID will not have a default value or a NULL value. If the primary key is deleted, DELETE CASCADE will be performed on the relation.

5) *user_id*: This attribute will serve as the Foreign Key. The purpose of this attribute is to link the expense to a user and will be an INT as its datatype. User ID will not have a default value or a NULL value. If the primary key is deleted, DELETE CASCADE will be performed on the relation.

F. Transactions Table

1) *transaction_id*: This attribute will serve as the Primary Key. The purpose of this attribute is to provide a unique ID for each transaction with INT as its datatype. Transaction ID will not have a default value or a NULL value; rather, it will start with 1 or the next available number that is not used in the relation.

2) *transaction_type*: This attribute will not serve as the Primary Key or the Foreign Key. The purpose of this attribute is to provide a type for the transaction, which will be a VARCHAR(50) as its datatype. Transaction Type will not have a default or NULL value.

3) *amount*: This attribute will not serve as the Primary Key or the Foreign Key. The purpose of this attribute is to provide the amount for the transaction, which will be a DECIMAL as its datatype. Amount will not have a default or NULL value.

4) *related_id*: This attribute will serve as the Foreign Key. The purpose of this attribute is to link which class (Assets, Expense, or Liability) the transaction belongs to, and it will be an INT as its datatype. Related ID will not have a default value or a NULL value. If the primary key is deleted, DELETE CASCADE will be performed on the relation.

5) *user_id*: This attribute will serve as the Foreign Key. The purpose of this attribute is to link the transaction to a user and will be an INT as its datatype. User ID will not have a default value or a NULL value. If the primary key is deleted, DELETE CASCADE will be performed on the relation.

G. Bookkeeping Table

1) *entry_id*: This attribute will serve as the Primary Key. The purpose of this attribute is to provide a unique ID for each bookkeeping entry with INT as its datatype. Entry ID will not have a default value or a NULL value; rather, it will start with 1 or the next available number that is not used in the relation.

2) *transaction_id*: This attribute will serve as the Foreign Key. The purpose of this attribute is to link which transaction the entry is related to and will be an INT as its datatype. Transaction ID will not have a default value or a NULL value. If the primary key is deleted, DELETE CASCADE will be performed on the relation.

3) *account_name*: This attribute will not serve as the Primary Key or the Foreign Key. The purpose of this attribute is to specify what type of account the transaction is placed in (Cash, Accounts Receivable, or Accounts Payable), which

will be a VARCHAR(50) as its datatype. Account Name will not have a default or NULL value.

4) *debit*: This attribute will not serve as the Primary Key or the Foreign Key. The purpose of this attribute is to record the debit amount in a transaction, which will be of type DECIMAL. Debit will not have a default value or allow NULL values.

5) *credit*: This attribute will not serve as the Primary Key or the Foreign Key. The purpose of this attribute is to record the credit amount in a transaction, which will be of type DECIMAL. Credit will not have a default value or allow NULL values.

6) *entry_date*: This attribute will not serve as the Primary Key or the Foreign Key. The purpose of this attribute is to store the date when the transaction entry occurred, which will be of type DATE. Entry Date will not have a default or NULL value.

7) *notes*: This attribute will not serve as the Primary Key or the Foreign Key. The purpose of this attribute is to store any relevant notes or descriptions related to the transaction, which will be a VARCHAR type. If notes are not provided, they can be NULL.

H. Audit Logs Table

1) *log_id*: This attribute will serve as the Primary Key. The purpose of this attribute is to uniquely identify each audit log entry. It will be of type INT and will auto-increment. Log ID will not have a default or NULL value.

2) *user_id*: This attribute will serve as the Foreign Key. The purpose of this attribute is to link which user performed the logged action and will be an INT as its datatype. User ID will not have a default value or a NULL value. If the primary key in the Users table is deleted, DELETE CASCADE will be performed on this relation.

3) *action_type*: This attribute will not serve as the Primary Key or the Foreign Key. The purpose of this attribute is to indicate the type of action performed (INSERT, UPDATE, DELETE, APPROVE, etc.), which will be a VARCHAR type. Action Type will not have a default or NULL value.

4) *table_name*: This attribute will not serve as the Primary Key or the Foreign Key. The purpose of this attribute is to specify the name of the table affected by the action, which will be a VARCHAR type. Table Name will not have a default or NULL value.

5) *record_id*: This attribute will not serve as the Primary Key or the Foreign Key. The purpose of this attribute is to identify the specific record in the affected table. It will be of type INT and will not have a default or NULL value.

6) *timestamp*: This attribute will not serve as the Primary Key or the Foreign Key. The purpose of this attribute is to record the exact date and time when the action occurred, which will be a DATETIME type. Timestamp will not have a default or NULL value.

7) *notes*: This attribute will not serve as the Primary Key or the Foreign Key. The purpose of this attribute is to store any additional comments or information about the logged action,

which will be a VARCHAR type. If notes are not provided, they can be NULL.

VI. FUNCTIONAL DEPENDENCIES

A. Departments

FD: department_id $\rightarrow$ department_name, manager_id
Each department is uniquely identified by department_id, which determines all other attributes. In BCNF, since department_id is a candidate key.

B. Users

FD: user_id $\rightarrow$ first_name, last_name, email, department_id, role; email $\rightarrow$ user_id
Each user is uniquely identified by user_id, which determines all other attributes. In BCNF, since user_id determinants are candidate keys.

C. Assets

FD: asset_id $\rightarrow$ asset_name, asset_type, purchase_date, value, department_id
Each asset is uniquely identified by asset_id, which determines all other attributes. In BCNF, since asset_id is a candidate key.

D. Liabilities

FD: liability_id $\rightarrow$ liability_name, amount, due_date, department_id
Each liability is uniquely identified by liability_id, which determines all other attributes. In BCNF, since liability_id is a candidate key.

E. Expenses

FD: expense_id $\rightarrow$ expense_name, amount, department_id, user_id
Each expense is uniquely identified by expense_id, which determines all other attributes. In BCNF, since expense_id is a candidate key.

F. Transactions

FD: transaction_id $\rightarrow$ transaction_type, amount, related_id, user_id
Each transaction is uniquely identified by transaction_id, which determines all other attributes. In BCNF, since transaction_id is a candidate key.

G. Bookkeeping

FD: entry_id $\rightarrow$ transaction_id, account_name, debit, credit, entry_date, notes
Each bookkeeping entry is uniquely identified by entry_id, which determines all other attributes. In BCNF, since entry_id is a candidate key.

H. Audit Logs

FD: log_id $\rightarrow$ user_id, action_type, table_name, record_id, timestamp, notes
Each audit log entry is uniquely identified by log_id, which determines all other attributes. In BCNF, since log_id is a candidate key.

VII. PROJECT PLAN

We have successfully developed the overall design plan for our project, which will include a total of 8 tables. Each table will contain at least 3000 rows of data. Since publicly available accounting datasets did not meet our specific design requirements, we plan to generate the data ourselves. Also we plan on using python as our programming language for the project and use POSTGRES SQL as our database server.